Module 5: Rise and fall of the clockwork universe

5.1 Models and rules

This module builds upon the work covered earlier in the course. The first section (**5.1.1**) uses simple techniques to model radioactive decay, capacitor charging and discharging and simple harmonic motion. In this framework, the formalism of the differential equation is developed along with the concept of field. There are many opportunities for practical work and empirical data can be compared and contrasted to the predictions made by the simple mathematical models. The field model is developed through consideration of gravitational fields.

The section raises questions about simplification in models, their usefulness and limitations (HSW7, HSW11). The difference can be highlighted between those models in which well-determined behaviour is due to exact rules operating on variables (as in the harmonic oscillator) and those in which well-determined behaviour is due to smooth averages over many particles (as in radioactive decay).

The second section (**5.1.2**) develops ideas about gravitational field strength and potential. Space flight and astronomical data can provide a context and there are further opportunities to consider the development of the modern view of the universe (HSW1, HSW6, HSW7, HSW11).

The third section (**5.1.3**) covers a descriptive and mainly qualitative outline of the main features of the observable universe consistent with the hot big bang model of its origin. The ideas of the universality of the speed of light and the relativistic consequence of time dilation are introduced.